ACCIOJOB • PROJECT PRESENTATION

OLYMPIC GAMES DATA ANALYSIS

Historical trends in sports, participants, medals and regional representation

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Rough presentation • Project plan and dashboard concept

Project objective

What the project is intended to answer

Explore historical Olympic data to identify trends and patterns across Games editions, sports, athletes, medals and regional representation.

Understand history

Compare Olympic editions, seasons, host cities and changes over time.

Explore competition

Study sports, events, participation and athlete demographics.

Communicate insights

Use Excel summaries and Power BI dashboards to present findings.

Outcome: a documented analytical workflow, an Excel workbook with 18 EDA result sheets, and a five-page Power BI report.

Problem statement

Why analyze Olympic data?

- Olympic records span many years and cover different entities, including Games, sports, events, people, medals and regions.
- These records need to be connected and summarized to make historical comparisons meaningful.
- The project investigates changes in participation, event scale, demographics, medal outcomes and regional representation.
- The analysis is descriptive: it highlights patterns in the supplied dataset rather than proving causes.

Core question

What patterns can the historical Olympic dataset reveal, and how can those patterns be communicated clearly through analysis and visualization?

Dataset overview

Relational database • PostgreSQL • public schema

11 interconnected tables

- Games and host locations
- Sports and events
- People and Games participation
- Competitor-event and medal records
- NOC codes and regional associations

51

Olympic editions

66

Sports

757

Events

128,854

People records

Data dictionary

Documents table purpose, key fields and relationships.

Counts are from the supplied database description; record counts represent different units of analysis.

Tools and technologies

Each tool supports a different stage of the workflow

PostgreSQL + SQL

Stores structured data; SQL joins and aggregates records to answer analytical questions.

Python + SQLAlchemy

Connects to PostgreSQL and runs queries from the Python environment.

Pandas

Receives query results as DataFrames for inspection and structured handling.

Microsoft Excel

Organizes the 18 query outputs and supporting summary sheets in one workbook.

Power BI Desktop

Connects the tables, applies model relationships and presents the analysis visually.

Workflow: PostgreSQL queries → Pandas DataFrames → Excel results → Power BI dashboard.

Database structure and relationships

ER diagram summarizes how the 11 tables connect

Main relationship paths

- Games ↔ City through the games_city bridge table
- Games ↔ Person through games_competitor
- Games competitor ↔ Event through competitor_event
- Sport → Event
- Person ↔ NOC region through person_region

Why this matters

The ER structure identifies the keys required to join tables for analysis. It also helps prevent treating participation records, unique people and medal-bearing event records as if they were the same measure.

The detailed ER diagram is included in the project documentation; add the same diagram to this slide if the rough deck needs a visual model.

Analysis methodology

From database inspection to dashboard reporting

01 **Connect** Connect Python to PostgreSQL with SQLAlchemy.

02 **Inspect** Review sample rows, columns and table relationships.

03 **Query** Run 18 SQL analyses across historical and participation questions.

04 **Organize** Read query outputs into Pandas DataFrames and organize results in Excel.

05 **Visualize** Build five Power BI pages using the connected data model.

Pandas was used for structured handling and inspection of SQL outputs—not as a replacement for SQL aggregation.

EDA plan: 18 analytical questions

Question groups and the kind of insight they target

Games history

Editions, seasons, host cities, decade patterns

Sports & events

Sport participation, event counts, recent and last-observed events

Athlete demographics

Gender categories, age, height and weight

Medal analysis

Medal-bearing records, medal-entry rates, repeat medal presence

Regional representation

Participants by region, regional changes, sport-specific patterns

The Excel workbook contains one result sheet per question, plus summary, data dictionary and supporting sheets.

Proposed dashboard layout

Five report pages group related questions for easier exploration

1 Olympic Overview

Editions by season, host cities, decades

2 Sports & Events

Sport counts, event categories, changes by edition

3 Athlete Demographics

Regional participants, gender entries, average age

4 Medal Analysis

Medal records by edition, sport and NOC

5 Regional Participation

Regions over time, participants and medal categories

Final presentation will use screenshots of the completed Power BI report rather than reproducing every chart on a separate slide.

Expected outcomes and scope

What the project delivers—and how findings should be read

Deliverables

18 SQL EDA outputs • Excel workbook • ER diagram • five-page Power BI report • project documentation.

Analytical outcomes

Historical comparisons, sport/event patterns, demographic summaries, medal-record analysis and regional participation views.

Interpretation guardrails

- Participant entries are not always unique individuals.
- Medal-bearing competitor-event records are not always unique medals.
- Historical region labels and event classifications need context.
- The project describes patterns; it does not establish causal explanations.

Rough deck • Prepared as the initial presentation plan; final deck will emphasize validated findings and dashboard screenshots.